Week 2 practice problems(Definite Integrals using the limit definition)

Question breakdowns:

Q1: Constant such as 5, 6, etc.

Q2: Variable such as 3x, 6x, etc.

Q3: Variable squared such as x^2 , $y^2, etc.$

Q4: Linear equation such as $2x + 3$,$5y - 3, etc.$

Q5: Quadratic equation such as $6 - x^2, etc.$

Q6: Quadratic equation such as $3x^2 + 2, etc.$ A boundary is negative and the $3x^2$ has a coefficient.

Q7: Quadratic equation such as $3x^2 + 7x, etc.$ The x^2 and x have coefficients.

Q8: Quadratic equation such as $x^2 - x + 3, etc.$

Q9: Quadratic equation such as $4x^2 + x + 2, etc.$ The x^2 has a coefficient, and the quadratic equation is broken down.

Q10: Cubic function such as $x^3 - 3x^2, etc.$

Q11: Cubic function such as $2x - x^3, etc.$

Q12: ?????

• Practice Questions

1. $\int_0^4 5\,dx$

$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\cdot \Delta x$

$= \lim_{n\to\infty} \sum_{i=1}^{n} 5 \cdot \frac{4}{n}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{20}{n} = \lim_{n\to\infty} \frac{20}{n} \sum_{i=1}^{n} 1 = \lim_{n\to\infty} \frac{20}{n}\cdot(n) = \lim_{n\to\infty} 20 = 20 \rightarrow$

$\Delta x = \frac{b-a}{n} = \frac{4-0}{n} = \frac{4}{n}$

$x_i = a + \Delta x \cdot i = 0 + \frac{4}{n}\cdot i = \frac{4i}{n}$

$x_i = 5$

2. $\int_0^2 3x\,dx$

$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\cdot \Delta x$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{6i}{n}\cdot\frac{2}{n}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{12i}{n^2} = \lim_{n\to\infty}\left[\frac{12}{n^2}\sum_{i=1}^{n} i\right] = \lim_{n\to\infty}\left[\frac{12^{6}}{n^2}\cdot\frac{n(n+1)}{2}\right]$

$= \lim_{n\to\infty} \frac{6n(n+1)}{n^2} = \lim_{n\to\infty} 6\left[\frac{n(n+1)}{n^2}\right] = \lim_{n\to\infty} 6\left[\frac{n}{n}+\frac{1}{n}\right]$

$= \lim_{n\to\infty} 6\left[1+\frac{1}{n}\right] = 6\left[1+\cancel{\frac{1}{\infty}}0\right] = 6 \rightarrow$

$\Delta x = \frac{b-a}{n} = \frac{2-0}{n} = \frac{2}{n}$

$x_i = a + \Delta x\, i = 0 + \frac{2}{n}\cdot i = \frac{2i}{n}$

$f(x_i) = 3\left(\frac{2i}{n}\right) = \frac{6i}{n}$

3. $\int_0^3 x^2\,dx$

$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{9i^2}{n^2}\cdot\frac{3}{n}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{27i^2}{n^3} = \lim_{n\to\infty} \frac{27}{n^3}\sum_{i=1}^{n} i^2 = \lim_{n\to\infty}\left[\frac{27^{9}}{n^3}\cdot\frac{n(n+1)(2n+1)}{\cancel{6}2}\right]$

$= \lim_{n\to\infty}\left[\frac{9n(n+1)(2n+1)}{2n^{\cancel{3}2}}\right] = \lim_{n\to\infty} 9\left[\frac{(n+1)(2n+1)}{2n^2}\right] = \lim_{n\to\infty} 9\left[\frac{2n^2+n+2n+1}{2n^2}\right]$

$= \lim_{n\to\infty} 9\left[\frac{2n^2}{2n^2}+\frac{3n}{2n^2}+\frac{1}{2n^2}\right] = \lim_{n\to\infty} 9\left[1+\frac{3}{2n}+\frac{1}{2n^2}\right] = \frac{9}{1}\left[1+0+0\right]$

$= 9 \rightarrow$

$$\lim_{n\to\infty} \frac{9}{2}\left[\frac{2n^2+3n+1}{n^2}\right] = \lim_{n\to\infty} \frac{9}{2}\left[\frac{2n^2}{n^2} + \frac{3n}{n^2} + \frac{1}{n^2}\right]$$

$$= \lim_{n\to\infty} \frac{9}{2}\left[2 + \frac{3}{n} + \frac{1}{n^2}\right] = \frac{9}{2}[2+0+0] = 9 \longrightarrow$$

4. $\int_0^1 (2x+3)\,dx$

$\Delta x = \frac{b-a}{n} = \frac{1-0}{n} = \frac{1}{n}$

$x_i = a + \Delta x_i = 0 + \frac{1}{n}\cdot i = \frac{i}{n}$

$f(x_i) = 2\left(\frac{i}{n}\right) + 3 = \frac{2i}{n} + 3$

$A: \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\cdot \Delta x$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{2i}{n} + 3\right]\cdot\frac{1}{n}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{2i}{n^2} + \frac{3}{n}\right] = \lim_{n\to\infty}\left[\frac{2}{n^2}\sum_{i=1}^{n} i + \frac{3}{n}\sum_{i=1}^{n} 1\right] = \lim_{n\to\infty}\left[\frac{2}{n^2}\cdot\frac{n(n+1)}{2} + \frac{3}{n}\cdot n\right]$

$= \lim_{n\to\infty} \frac{n(n+1)}{n^2} + 3 = \lim_{n\to\infty} \frac{n+1}{n} + 3 = \lim_{n\to\infty} \frac{n}{n} + \frac{1}{n} + 3$

$= \lim_{n\to\infty} 1 + \frac{1}{n} + 3 = 1 + 0 + 3 = 4 \longrightarrow$

5. $\int_0^4 (6 - x^2)\,dx$

$\Delta x = \frac{b-a}{n} = \frac{4-0}{n} = \frac{4}{n}$

$x_i = a + \Delta x_i = 0 + \frac{4}{n}\cdot i = \frac{4i}{n}$

$f(x_i) = 6 - \left(\frac{4i}{n}\right)^2 = 6 - \frac{16i^2}{n^2}$

$A: \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\cdot \Delta x$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[6 - \frac{16i^2}{n^2}\right]\cdot\frac{4}{n}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{24}{n} - \frac{64i^2}{n^3}\right] = \lim_{n\to\infty}\left[\frac{24}{n}\sum_{i=1}^{n} 1 - \frac{64}{n^3}\sum_{i=1}^{n} i^2\right]$

$= \lim_{n\to\infty}\left[\frac{24}{n}\cdot n - \frac{64}{n^3}\cdot\frac{n(n+1)(2n+1)}{6}\right] \lim_{n\to\infty}\left[24 - \frac{32n(n+1)(2n+1)}{3n^3}\right]$

→ Attached = can't × 24 and $\frac{32}{3}$

$= \lim_{n\to\infty} 24 - \frac{32}{3}\left[\frac{(n+1)(2n+1)}{n^2}\right] = \lim_{n\to\infty} \frac{40}{3}\left[\frac{2n^2+3n+1}{n^2}\right]$

$= \lim_{n\to\infty} \frac{40}{3}\left[\frac{2n^2}{n^2} + \frac{3n}{n^2} + \frac{1}{n^2}\right] = \lim_{n\to\infty} \frac{40}{3}\left[2 + \frac{3}{n} + \frac{1}{n^2}\right] = \lim_{n\to\infty}$

$= \frac{40}{3}[2+0+0]$

$$= \lim_{n\to\infty} 24 - \frac{32}{3}\left[\frac{2n^2}{n^2} + \frac{3n}{n^2} + \frac{1}{n^2}\right] = \lim_{n\to\infty} 24 - \frac{32}{3}\left[2 + \frac{3}{n} + \frac{1}{n^2}\right]$$

$$= 24 - \frac{32}{3}[2+0+0] = 24 - \frac{32}{3}\cdot 2 = 24 - \frac{64}{3} = \frac{8}{3}$$

i. $\int_{-2}^{1} (3x^2+2)\,dx$

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\cdot \Delta x$$

$$\Delta x = \frac{b-a}{n} = \frac{1-(-2)}{n} = \frac{3}{n}$$

$$x_i = a + \Delta x i = -2 + \frac{3}{n} i = -2 + \frac{3i}{n}$$

$$f(x_i) = 3\left(-2 + \frac{3i}{n}\right)^2 + 2$$

$$= 3\left(4 + \frac{9i^2}{n^2}\right) + 2 \qquad F^2 + 2FL + L^2$$

$$= 12 + \frac{27i^2}{n^2} + 2 \quad \text{messed up the entire calculation}$$

$$= 14 + \frac{27i^2}{n^2}$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[14 + \frac{27i^2}{n^2}\right]\cdot \frac{3}{n}$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{42}{n} + \frac{81i^2}{n^3}$$

$$= \lim_{n\to\infty} \left[\frac{42}{n}\sum_{i=1}^{n} + \frac{81}{n^3}\sum_{i=1}^{n} i^2\right] = \lim_{n\to\infty} \left[\frac{42}{n}\cdot n + \frac{81}{n^3}\cdot \frac{n(n+1)(2n+1)}{6}\right]$$

$$= \lim_{n\to\infty} 42 + \left[\frac{27n(n+1)(2n+1)}{2n^3}\right] = \lim_{n\to\infty} 42 + \frac{27}{2}\left[\frac{2n^2+3n+1}{n^2}\right]$$

$$= \lim_{n\to\infty} 42 + \frac{27}{2}\left[\frac{2n^2}{n^2} + \frac{3n}{n^2} + \frac{1}{n^2}\right] = \lim_{n\to\infty} 42 + \frac{27}{2}\left[2 + \frac{3}{n} + \frac{1}{n^2}\right]$$

$$= 42 +$$

i. $\int_{-2}^{1} (3x^2+2)\,dx$

$$\Delta x = \frac{b-a}{n} = \frac{1-(-2)}{n} = \frac{3}{n}$$

$$x_i = a + \Delta x i = -2 + \frac{3}{n} i = -2 + \frac{3i}{n}$$

$$f(x_i) = 3\left(-2 + \frac{3i}{n}\right)^2 + 2$$

$$= 3\left[4 - \frac{6i}{n} + \frac{9i^2}{n^2}\right] + 2$$

$$= 12 - \frac{18i}{n} + \frac{27i^2}{n^2} + 2$$

$$= 14 - \frac{18i}{n} + \frac{27i^2}{n^2}$$

$$F^2 + 2FL + L^2$$

6. $\int_{-2}^{1}(3x^2+2)\,dx$

$A=\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i)\cdot\Delta x$

$\Delta x=\frac{b-a}{n}=\frac{1-(-2)}{n}=\frac{3}{n}$

$x_i=a+\Delta x_i=-2+\frac{3}{n}i=-2+\frac{3i}{n}$

$f(x_i)=3\left(-2+\frac{3i}{n}\right)^2+2$

$=3\left[4-\frac{12i}{n}+\frac{9i^2}{n^2}\right]+2$

$=12-\frac{36i}{n}+\frac{27i^2}{n^2}+2$

$=14-\frac{36i}{n}+\frac{27i^2}{n^2}$

$=\lim_{n\to\infty}\sum_{i=1}^{n}\left[\frac{42}{n}-\frac{108i}{n^2}+\frac{81i^2}{n^2}\right]$

$=\lim_{n\to\infty}\left[\frac{42}{n}\sum_{i=1}^{n}1-\frac{108}{n^2}\sum_{i=1}^{n}i+\frac{81}{n^2}\sum_{i=1}^{n}i^2\right]$ $\frac{n(n+1)}{2}$ $\frac{n(n+1)(2n+1)}{6}$ $\frac{108}{n}\sum_{i=1}^{n}i+\frac{81}{n}\sum_{i=1}^{n}i^2$

$=\lim_{n\to\infty}\left[\frac{42}{n}\cdot n-\frac{108}{n^2}\cdot\frac{n(n+1)}{2}+\frac{81}{n^2}\cdot\frac{n(n+1)(2n+1)}{6}\right]$

▷ 6 not 2

$=\lim_{n\to\infty}\left[42-\frac{54n(n+1)}{n}+\frac{81n(n+1)(2n+1)}{2n^2}\right]$ $\frac{81}{6}=\frac{27}{2}$

$=\lim_{n\to\infty}\left[42-54\left(\frac{n(n+1)}{n}\right)+\frac{81}{2}\left(\frac{(n+1)(2n+1)}{n^2}\right)\right]$

$=\lim_{n\to\infty}42-54\left(\frac{n^2+1}{n}\right)+\frac{81}{2}\left(\frac{2n^2+3n+1}{n^2}\right)$

$=\lim_{n\to\infty}42-54\left(\frac{n^2}{n}+\frac{1}{n}\right)+\frac{81}{2}\left(\frac{2n^2}{n^2}+\frac{3n}{n^2}+\frac{1}{n^2}\right)$

$=\lim_{n\to\infty}42-54\left(n+\frac{1}{n}\right)+\frac{81}{2}\left(2+\frac{3}{n}+\frac{1}{n^2}\right)$

$=42-54(0+0)$

$=\lim_{n\to\infty}\left[42-\frac{54(n+1)}{n}+\frac{81n(n+1)(2n+1)}{2n^2}\right]$

$=\lim_{n\to\infty}\left[42-54\left(\frac{n}{n}+\frac{1}{n}\right)+\frac{81}{2}\left(\frac{2n^2+3n+1}{n^2}\right)\right.$

$=\lim_{n\to\infty}\left[42-54\left(1+\frac{1}{n}\right)+\frac{81}{2}\left(\frac{2n^2}{n^2}+\frac{3n}{n^2}+\frac{1}{n^2}\right)\right]$

$\lim = 42 - 54(1+0) + \frac{27}{2}(2+0+0) = 42 - 54 + 27 = 15$

7 $\int_1^5 (3x^2+7x)\,dx$

$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x)\Delta x$

$\Delta x = \frac{b-a}{n} = \frac{5-1}{n} = \frac{4}{n}$

$x_i = a + \Delta x\, i = 1 + \frac{4}{n} i = 1 + \frac{4i}{n}$

$f(x_i) = 3\left(1+\frac{4i}{n}\right)^2 + 7\left(1+\frac{4i}{n}\right)$

$= 3\left[1 + \frac{8i}{n} + \frac{16i^2}{n^2}\right] + 7 + \frac{28i}{n}$

$= 3 + \frac{24i}{n} + \frac{16i^2}{n^2} + 7 + \frac{28i}{n}$

$= 10 + \frac{52i}{n} + \frac{16i^2}{n^2}$

didn't x 3 into the 16

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[10 + \frac{52i}{n} + \frac{16i^2}{n^2}\right] \cdot \frac{4}{n}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{40}{n} + \frac{208i}{n^2} + \frac{64i^2}{n^3}\right]$

$= \lim_{n\to\infty} \left[\frac{40}{n}\sum_{i=1}^{n} 1 + \frac{208}{n^2}\sum_{i=1}^{n} i + \frac{64}{n^3}\sum_{i=1}^{n} i^2\right]$

$= \lim_{n\to\infty} \left[\frac{40}{n}\cdot n + \frac{208}{n^2}\cdot\frac{n(n+1)}{2} + \frac{64}{n^3}\cdot\frac{n(n+1)(2n+1)}{6}\right]$

$= \lim_{n\to\infty} \left[40 + \frac{104n(n+1)}{n^2} + \frac{32n(n+1)(2n+1)}{3n^3}\right]$

$= \lim_{n\to\infty} \left[40 + 104\left(\frac{n}{n}+\frac{1}{n}\right) + \frac{32}{3}\left(\frac{2n^2+3n+1}{n^2}\right)\right]$

$= \lim_{n\to\infty} \left[40 + 104\left(1+\frac{1}{n}\right) + \frac{32}{3}\left(2 + \frac{3n}{n^2} + \frac{1}{n^2}\right)\right]$

$= 40 + 104(1+0) + \frac{32}{3}(2+0+0) = 40 + 104 + \frac{64}{3} = \frac{496}{3}$

7. $\int_1^5 (3x^2+7x)\,dx$

$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x$

$\Delta x = \frac{b-a}{n} = \frac{5-1}{n} = \frac{4}{n}$

$x_i = a + \Delta x\, i = 1 + \frac{4}{n} i = \frac{4i}{n+1}$

$f(x_i) = 3\left[1+\frac{4i}{n}\right]^2 + 7\left[1+\frac{4i}{n}\right]$

$= 3\left[1 + \frac{8i}{n} + \frac{16i^2}{n^2}\right] + 7 + \frac{28i}{n}$

$= 3 + \frac{24i}{n} + \frac{48i^2}{n^2} + 7 + \frac{28i}{n}$

$= 10 + \frac{52i}{n} + \frac{48i^2}{n^2}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[10 + \frac{52i}{n} + \frac{48i^2}{n^2}\right] \cdot \frac{4}{n}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{40}{n} + \frac{208i}{n^2} + \frac{192i^2}{n^3}\right]$

$= \lim_{n\to\infty} \left[\frac{40}{n}\sum_{i=1}^{n} 1 + \frac{208}{n^2}\sum_{i=1}^{n} i + \frac{192}{n^3}\sum_{i=1}^{n} i^2\right]$

$= \lim_{n\to\infty} \frac{40}{n}\cdot n + \frac{208}{n^2}\cdot\frac{n(n+1)}{2} + \frac{192}{n^3}\cdot\frac{n(n+1)(2n+1)}{6}$

$$= \lim_{n\to\infty} 40 + \frac{104n(n+1)}{n^2} + \frac{32n(n+1)(2n+1)}{n^3}$$

$$= \lim_{n\to\infty} 40 + 104\left[\frac{n}{n} + \frac{1}{n}\right] + 32\left[\frac{2n^2+3n+1}{n^2}\right]$$

$$= \lim_{n\to\infty} 40 + 104\left[1 + \frac{1}{n}\right] + 32\left[\frac{2n^2}{n^2} + \frac{3n}{n^2} + \frac{1}{n^2}\right]$$

$$= 40 + 104[1+0] + 32[2+0+0] = 40+104+64 = 208$$

8.) $\int_0^1 (x^2 - x + 3)\,dx$

$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$

$\Delta x = \frac{b-a}{n} = \frac{1-0}{n} = \frac{1}{n}$

$x_i = a + \Delta x\, i = 0 + \frac{1}{n} i = \frac{i}{n}$

$f(x_i) = \left(\frac{i}{n}\right)^2 - \left(\frac{i}{n}\right) + 3$

$= \frac{i^2}{n^2} - \frac{i}{n} + 3$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{i^2}{n^2} - \frac{i}{n} + 3\right]\cdot\frac{1}{n}$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{i^2}{n^3} - \frac{i}{n^2} + \frac{3}{n}\right]$$

$$= \lim_{n\to\infty} \left[\frac{1}{n^3}\sum_{i=1}^{n} i^2 - \frac{1}{n^2}\sum_{i=1}^{n} i + \frac{3}{n}\sum_{i=1}^{n} 1\right]$$

$$= \lim_{n\to\infty} \left[\frac{1}{n^3}\cdot\frac{n(n+1)(2n+1)}{6} - \frac{1}{n^2}\cdot\frac{n(n+1)}{2} + \frac{3}{n}\cdot n\right]$$

$$= \lim_{n\to\infty} \frac{n(n+1)(2n+1)}{6n^2} - \frac{n(n+1)}{2n^2} + 3$$

$$= \lim_{n\to\infty} \frac{1}{6}\left[\frac{2n^2+3n+1}{n^2}\right] - \frac{1}{2}\left[\frac{n+1}{n}\right] + 3$$

$$= \lim_{n\to\infty} \frac{1}{6}\left[\frac{2n^2}{n^2} + \frac{3n}{n^2} + \frac{1}{n^2}\right] - \frac{1}{2}\left[\frac{n}{n} + \frac{1}{n}\right] + 3$$

$$= \frac{1}{6}(2+0+0) - \frac{1}{2}(1+0) + 3 = \frac{1}{3} - \frac{1}{2} + 3 = \frac{17}{6}$$

~~9.) $\int_{-1}^{2} (4x^2 + x + 2)\,dx$~~

~~$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$~~

~~$\Delta x = \frac{b-a}{n} = \frac{2-(-1)}{n} = \frac{3}{n}$~~

~~$x_i = a + \Delta x\, i = -1 + \frac{3}{n} i = -1 + \frac{3i}{n}$~~

~~$f(x_i) = 4\left(-1 + \frac{3i}{n}\right) + \left(-1 + \frac{3i}{n}\right) + 2$~~

~~$= -4 + \frac{12i}{n} - 1 + \frac{3i}{n} + 2$~~

~~$= -3 + \frac{15i}{n}$~~

~~$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[-3 + \frac{15i}{n}\right]\cdot\frac{3}{n}$~~

9. $\int_{-1}^{2}(4x^2+x+2)\,dx$

$\Delta x = \frac{b-a}{n} = \frac{2-(-1)}{n} = \frac{3}{n}$

$x_i = a + \Delta x\, i = -1 + \frac{3}{n} i = -1 + \frac{3i}{n}$

$f(x_i) = 4\left(-1+\frac{3i}{n}\right)^2 + \left(-1+\frac{3i}{n}\right) + 2$

$= 4\left(1 - \frac{6i}{n} + \frac{9i^2}{n^2}\right) + \left(-1+\frac{3i}{n}\right) + 2$

$= 4 - \frac{24i}{n} + \frac{36i^2}{n^2} - 1 + \frac{3i}{n} + 2$

$= 6 - \frac{21i}{n} + \frac{36i^2}{n^2} - 1$

$= 5 - \frac{21i}{n} + \frac{36i^2}{n^2}$

$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[5 - \frac{21i}{n} + \frac{36i^2}{n^2}\right] \cdot \frac{3}{n}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{15}{n} - \frac{63i}{n^2} + \frac{108i^2}{n^3}\right]$

$= \lim_{n\to\infty} \frac{15}{n}\sum_{i=1}^{n} 1 - \frac{63}{n^2}\sum_{i=1}^{n} i + \frac{108}{n^3}\sum_{i=1}^{n} i^2$

$= \lim_{n\to\infty} \left[\frac{15}{n}\cdot n - \frac{63}{n^2}\cdot\frac{n(n+1)}{2} + \frac{108}{n^3}\cdot\frac{n(n+1)(2n+1)}{6}\right]$

$= \lim_{n\to\infty} \left[15 - \frac{63n(n+1)}{2n^2} + \frac{18n(n+1)(2n+1)}{n^3}\right]$

$= \lim_{n\to\infty} \left[15 - \frac{63}{2}\cdot\frac{n+1}{n} + 18\cdot\frac{2n^2+3n+1}{n^2}\right]$

$= \lim_{n\to\infty} \left[15 - \frac{63}{2}\cdot\left[\frac{n}{n}+\frac{1}{n}\right] + 18\cdot\left[\frac{2n^2}{n^2}+\frac{3n}{n^2}+\frac{1}{n^2}\right]\right]$

$= 15 - \frac{63}{2}[1+0] + 18[2+0+0] = 15 - \frac{63}{2} + 36 = \frac{39}{2}$

10.) $\int_0^1 (x^3 - 3x^2)\,dx$

$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x$

$\Delta x = \frac{b-a}{n} = \frac{1-0}{n} = \frac{1}{n}$

$x_i = a + \Delta x\, i = 0 + \frac{1}{n} i = \frac{i}{n}$

$f(x_i) = \left(\frac{i}{n}\right)^3 - 3\left(\frac{i}{n}\right)^2$

$= \frac{i^3}{n^3} - 3\left(\frac{i^2}{n^2}\right)$

$= \frac{i^3}{n^3} - \frac{3i^2}{n^2}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{i^3}{n^3} - \frac{3i^2}{n^2}\right] \cdot \frac{1}{n}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{i^3}{n^4} - \frac{3i^2}{n^3}\right]$

$= \lim_{n\to\infty} \frac{1}{n^4} \sum_{i=1}^{n} i^3 - \frac{3}{n^3} \sum_{i=1}^{n} i^2$

$= \lim_{n\to\infty} \left[\frac{1}{n^4} \cdot \left[\frac{n^2(n+1)^2}{4}\right] - \frac{3}{n^3} \cdot \left[\frac{n(n+1)(2n+1)}{6}\right]\right]$

$= \lim_{n\to\infty} \left[\frac{n^2(n+1)^2}{4n^4} - \frac{n(n+1)(2n+1)}{2n^3}\right]$

$= \lim_{n\to\infty} \left[\frac{(n+1)^2}{4n^2} - \frac{(n+1)(2n+1)}{n^2}\right]$

$= \lim_{n\to\infty} \frac{1}{4} \cdot \frac{(n+1)^2}{n^2} - \frac{2n^2+3n+1}{n^2} \cdot \lim_{n\to\infty} \frac{1}{4} \cdot \frac{n^2+2n+1}{n^2} - \frac{2n^2+3n+1}{n^2}$

$= \lim_{n\to\infty} \frac{1}{4}\left[\frac{n^2}{n^2} + \frac{2n}{n^2} + \frac{1}{n^2}\right] - \left[\frac{2n^2}{n^2} + \frac{3n}{n^2} + \frac{1}{n^2}\right]$

$= \lim_{n\to\infty} \frac{1}{4}[1 + 0 + 0] - [2 + 0 + 0]$

$= \frac{1}{4} - 2 = -\frac{7}{4}$

10.) $\int_0^1 (x^3 - 3x^2)\,dx$

$\Delta x = \frac{b-a}{n} = \frac{1-0}{n} = \frac{1}{n}$

$x_i = a + \Delta x_i = 0 + \frac{1}{n} i = \frac{i}{n}$

$f(x_i) = \left(\frac{i}{n}\right)^3 - 3\left(\frac{i}{n}\right)^2$

$= \frac{i^3}{n^3} - 3\left(\frac{i^2}{n^2}\right)$

$= \frac{i^3}{n^3} - \frac{3i^2}{n^2}$

$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{i^3}{n^3} - \frac{3i^2}{n^2}\right] \cdot \frac{1}{n}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{i^3}{n^4} - \frac{3i^2}{n^3}\right]$

$= \lim_{n\to\infty} \frac{1}{n^4} \sum_{i=1}^{n} i^3 - \frac{3}{n^3} \sum_{i=1}^{n} i^2$

$= \lim_{n\to\infty} \left[\frac{1}{n^4} \cdot \left[\frac{n^2(n+1)^2}{4}\right] - \frac{3}{n^3} \cdot \left[\frac{n(n+1)(2n+1)}{6}\right]\right]$

$= \lim_{n\to\infty} \left[\frac{n^2(n+1)^2}{4n^4} - \frac{n(n+1)(2n+1)}{2n^3}\right]$

had forgotten the 2 before, but it carries down & cancels out in one of the last steps

$= \lim_{n\to\infty} \left[\frac{(n+1)^2}{4n^2} - \frac{(n+1)(2n+1)}{2n^2}\right]$

$= \lim_{n\to\infty} \frac{1}{4} \cdot \frac{(n+1)^2}{n^2} - \frac{2n^2+3n+1}{2n^2} \cdot \lim_{n\to\infty} \frac{1}{4} \cdot \frac{n^2+2n+1}{n^2} - \frac{2n^2+3n+1}{2n^2}$

$= \lim_{n\to\infty} \frac{1}{4}\left[\frac{n^2}{n^2} + \frac{2n}{n^2} + \frac{1}{n^2}\right] - \left[\frac{2n^2}{2n^2} + \frac{3n}{n^2} + \frac{1}{n^2}\right]$

$= \lim_{n\to\infty} \frac{1}{4}[1 + 0 + 0] \quad \frac{1+0+0}{2+0+0}$

$\frac{1}{4}\left[\frac{2}{2}\right] = \frac{1}{4}$ $\quad \frac{1}{4}[1+0+0] - [1+0+0] = -\frac{3}{4}$ →

11. $\int_0^2 (2x - x^3)\,dx$

$\Delta x = \frac{b-a}{n} = \frac{2-0}{n} = \frac{2}{n}$

$x_i = a + \Delta x_i = 0 + \frac{2}{n} i = \frac{2i}{n}$

$f(x_i) = 2\left(\frac{2i}{n}\right) - \left(\frac{2i}{n}\right)^3$

$= \frac{4i}{n} - \frac{8i^3}{n^3}$

$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{4i}{n} - \frac{8i^3}{n^3}\right] \cdot \frac{2}{n}$

$= \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{8i}{n^2} - \frac{16i^3}{n^4}\right]$

$$\lim_{n\to\infty} \frac{8}{n^2}\sum_{i=1}^{n} i - \frac{16}{n^4}\sum_{i=1}^{n} i^3$$

$$= \lim_{n\to\infty}\left[\frac{8}{n^2}\cdot\frac{4n(n+1)}{2} - \frac{16}{n^4}\cdot\frac{n^2(n+1)^2}{4}\right]$$

$$= \lim_{n\to\infty}\left[\frac{4n(n+1)}{n^2} - \frac{4n^2(n+1)^2}{n^4}\right]$$

$$= \lim_{n\to\infty} 4\cdot\frac{n+1}{n} - 4\cdot\frac{(n+1)^2}{n^2} = \lim_{n\to\infty} 4\left[\frac{n}{n}+\frac{1}{n}\right] - 4\left[\frac{n^2+2n+1}{n^2}\right]$$

$$= \lim_{n\to\infty} 4\left[\frac{n}{n}+\frac{1}{n}\right] - 4\left[\frac{n^2}{n^2}+\frac{2n}{n^2}+\frac{1}{n^2}\right]$$

$$= \lim_{n\to\infty} 4\left[1+\frac{1}{n}\right] - 4\left[1+\frac{2}{n}+\frac{1}{n^2}\right]$$

$$= 4[1+0] - 4[1+0+0] = 0 \rightarrow$$

12) $\int_0^1 \sqrt{1-x^2}$